

Convexity of the Real Hexablock

Automated mathematics run `fa_banach_001`, lane 2

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Status: full solution likely valid. The real hexablock is convex. The proof replaces the source’s complicated Hessian calculation by an exact radical formula for its boundary-height function and the elementary concavity of the geometric mean.

Abstract

Biswas, Pal, and Tomar conjectured that the intersection of the hexablock $\mathbb{H} \subset \mathbb{C}^4$ with $\mathbb{R}^4$ is convex. They reduced the conjecture to concavity of an explicitly defined height function K on the real tetrablock, but its nested radical formula made the Hessian appear intractable. We prove the exact identity

$$K(x_1, x_2, x_3) = \frac{1}{2} \left(\sqrt{(1+x_3)^2 - (x_1+x_2)^2} + \sqrt{(1-x_3)^2 - (x_1-x_2)^2} \right).$$

Equivalently, if $L_1, \dots, L_4$ are the four positive affine facet coordinates of the real tetrablock tetrahedron, then $K = \frac{1}{2}(\sqrt{L_2 L_3} + \sqrt{L_1 L_4})$. The geometric mean is concave on the positive quadrant, so K is concave and the conjecture follows.

1 Source question

The source is I. Biswas, S. Pal, and N. Tomar, *The Hexablock: a domain associated with the μ -synthesis in $M_2(\mathbb{C})$* , arXiv:2506.15149v2 (2026)[1]. On PDF page 38, the authors write: “We conjecture that the intersection of the hexablock with $\mathbb{R}^4$ is also convex.” Their Proposition 6.3 says that this is equivalent to concavity of the height function recalled below.

2 Definitions and proof intuition

The real tetrablock $\mathbb{E} \cap \mathbb{R}^3$ is the open tetrahedron with vertices

$$(1, 1, 1), \quad (1, -1, -1), \quad (-1, 1, -1), \quad (-1, -1, 1).$$

For $x = (x_1, x_2, p)$ in this tetrahedron, set

$$\beta_1 = \frac{x_1 - x_2 p}{1 - p^2}, \quad \beta_2 = \frac{x_2 - x_1 p}{1 - p^2}.$$

Then $|\beta_1| + |\beta_2| < 1$, and

$$x_1 = \beta_1 + p\beta_2, \quad x_2 = \beta_2 + p\beta_1. \tag{1}$$

where $z_1(x), z_2(x)$ are points in $(-1, 1)$ corresponding to $x = (x_1, x_2, x_3) \in \mathbb{E}$ as in Corollary 2.2.

Agler et al. [3] proved that $\mathbb{G}_2 \cap \mathbb{R}^2$ and $\mathbb{P} \cap \mathbb{R}^3$ are convex sets. An interested reader is referred to Section 9 in [3] for further details. A similar result holds for the tetrablock. In fact, Abouhajar et al. [2] proved that $\mathbb{E} \cap \mathbb{R}^3$ is an open tetrahedron and so, a convex set. We conjecture that the intersection of the hexablock with $\mathbb{R}^4$ is also convex. However, verifying this appears to be challenging at this point, mainly because of elaborated and complicated computations. We present an equivalent criterion for determining if $\mathbb{H} \cap \mathbb{R}^4$ is convex. To see this, we need a few terminologies from the literature.

Definition 6.2. Let $C \subseteq \mathbb{R}^n$ be a convex set. A function $f : C \rightarrow \mathbb{R}$ is said to be *concave* if

$$tf(x) + (1-t)f(y) \leq f(tx + (1-t)y)$$

for every $x, y \in C$ and $t \in (0, 1)$. Also, a function $f : C \rightarrow \mathbb{R}$ is said to be *convex* if

$$tf(x) + (1-t)f(y) \geq f(tx + (1-t)y)$$

for every $x, y \in C$ and $t \in (0, 1)$.

Proposition 6.3. The set $\mathbb{H} \cap \mathbb{R}^4$ is convex if and only if the map $K : \mathbb{E} \cap \mathbb{R}^3 \rightarrow \mathbb{R}$ given by

$$K(x) = \frac{1 - x_1 z_1(x) - x_2 z_2(x) + x_3 z_1(x) z_2(x)}{\sqrt{(1 - (z_1(x))^2)(1 - (z_2(x))^2)}}$$

is concave, where $z_1(x), z_2(x)$ are the points in the open interval $(-1, 1)$ corresponding to $x = (x_1, x_2, x_3) \in \mathbb{E}$ as in Corollary 2.2.

Proof. Assume that $\mathbb{H} \cap \mathbb{R}^4$ is convex. Let $x = (x_1, x_2, x_3), y = (y_1, y_2, y_3) \in \mathbb{E} \cap \mathbb{R}^3$ and let $t \in (0, 1)$. For $\varepsilon \in (0, 1)$, we define $a_\varepsilon(x) = \varepsilon K(x)$ and $b_\varepsilon(y) = \varepsilon K(y)$. By Proposition 6.1, we have that $(a_\varepsilon(x), x_1, x_2, x_3), (b_\varepsilon(y), y_1, y_2, y_3) \in \mathbb{H} \cap \mathbb{R}^4$, because $|a_\varepsilon(x)K(x)^{-1}| = |b_\varepsilon(y)K(y)^{-1}| = \varepsilon$. Since

Figure 1: The conjecture and exact concavity reduction in the source, PDF page 38.

The source's Corollary 2.2 defines the unique maximizing pair by

$$z_1 = \frac{2\beta_1}{1 + \beta_1^2 - \beta_2^2 + \sqrt{(1 + \beta_1^2 - \beta_2^2)^2 - 4\beta_1^2}}, \quad (2)$$

$$z_2 = \frac{2\beta_2}{1 + \beta_2^2 - \beta_1^2 + \sqrt{(1 + \beta_2^2 - \beta_1^2)^2 - 4\beta_2^2}}. \quad (3)$$

Its Proposition 6.3 uses the height

$$K(x) = \frac{1 - x_1 z_1 - x_2 z_2 + x_3 z_1 z_2}{\sqrt{(1 - z_1^2)(1 - z_2^2)}} \quad (4)$$

and proves that $\mathbb{H} \cap \mathbb{R}^4$ is convex if and only if K is concave on $\mathbb{E} \cap \mathbb{R}^3$.

Proof intuition. The nested radicals in (2)–(3) hide hyperbolic half-angle formulas. Put $u = \beta_1 + \beta_2$ and $v = \beta_1 - \beta_2$. The two discriminants become the same product $(1 - u^2)(1 - v^2)$, and the maximizing points become hyperbolic tangents of half-sums. Substitution into (4) makes all hyperbolic terms cancel, leaving two square roots. In tetrahedral facet coordinates, each is a geometric mean. Thus the apparent three-variable Hessian problem is really the concavity of $(a, b) \mapsto \sqrt{ab}$.

3 Main theorem

Theorem 1 (Real-hexablock convexity). *For every $x = (x_1, x_2, p) \in \mathbb{E} \cap \mathbb{R}^3$, the height in (4) has the closed form*

$$K(x) = \frac{1}{2} \left[\sqrt{(1+p)^2 - (x_1 + x_2)^2} + \sqrt{(1-p)^2 - (x_1 - x_2)^2} \right]. \quad (5)$$

It is concave on $\mathbb{E} \cap \mathbb{R}^3$. Consequently, $\mathbb{H} \cap \mathbb{R}^4$ is convex.

Proof. Put

$$u = \beta_1 + \beta_2, \quad v = \beta_1 - \beta_2.$$

The condition $|\beta_1| + |\beta_2| < 1$ is equivalent to $|u|, |v| < 1$. Let

$$U = \operatorname{artanh} u, \quad V = \operatorname{artanh} v, \quad A = \frac{U+V}{2}, \quad B = \frac{U-V}{2}.$$

Since $1 + \beta_1^2 - \beta_2^2 = 1 + uv$ and $2\beta_1 = u + v$,

$$(1 + \beta_1^2 - \beta_2^2)^2 - 4\beta_1^2 = (1 - u^2)(1 - v^2).$$

The analogous discriminant in (3) is the same. The hyperbolic half-angle identities therefore give

$$\begin{aligned} z_1 &= \frac{u+v}{1+uv+\sqrt{(1-u^2)(1-v^2)}} = \tanh A, \\ z_2 &= \frac{u-v}{1-uv+\sqrt{(1-u^2)(1-v^2)}} = \tanh B. \end{aligned}$$

Substitute these expressions into (4). Because $\sqrt{1-z_1^2} = \operatorname{sech} A$ and similarly for z_2 ,

$$\begin{aligned} K &= \cosh A \cosh B - x_1 \sinh A \cosh B - x_2 \cosh A \sinh B + p \sinh A \sinh B \\ &= \frac{1}{2} [(1+p) \cosh U + (1-p) \cosh V - (x_1 + x_2) \sinh U - (x_1 - x_2) \sinh V]. \end{aligned}$$

Equation (1) yields $x_1 + x_2 = (1+p)u$ and $x_1 - x_2 = (1-p)v$. Hence

$$\begin{aligned} K &= \frac{1}{2} [(1+p)(\cosh U - u \sinh U) + (1-p)(\cosh V - v \sinh V)] \\ &= \frac{1}{2} [(1+p) \operatorname{sech} U + (1-p) \operatorname{sech} V]. \end{aligned}$$

Since $\operatorname{sech}(\operatorname{artanh} t) = \sqrt{1-t^2}$, and since $u = (x_1 + x_2)/(1+p)$, $v = (x_1 - x_2)/(1-p)$, formula (5) follows.

It remains to prove concavity. Define the four affine facet functions

$$\begin{aligned} L_1 &= 1 - x_1 + x_2 - p, & L_2 &= 1 - x_1 - x_2 + p, \\ L_3 &= 1 + x_1 + x_2 + p, & L_4 &= 1 + x_1 - x_2 - p. \end{aligned}$$

They are positive precisely on the open tetrahedron, and

$$K = \frac{1}{2} \left(\sqrt{L_2 L_3} + \sqrt{L_1 L_4} \right). \quad (6)$$

For positive a, b, c, d and $t \in [0, 1]$,

$$\begin{aligned} (ta + (1-t)c)(tb + (1-t)d) - \left(t\sqrt{ab} + (1-t)\sqrt{cd}\right)^2 \\ = t(1-t)(\sqrt{ad} - \sqrt{bc})^2 \geq 0. \end{aligned}$$

Taking square roots proves concavity of $(a, b) \mapsto \sqrt{ab}$ on the positive quadrant. Composition with affine maps preserves concavity, so (6) is concave. Source Proposition 6.3 now implies that $\mathbb{H} \cap \mathbb{R}^4$ is convex. $\square$

4 Verification and scope

The reusable script `code/verify_closed_form.py` independently evaluates both the source formula and (5). With seed 250615149, it compared them on 100,000 random interior points; the maximum absolute discrepancy was 4.17×10^{-14} . It also tested 100,000 random Jensen triples and found no negative concavity gap beyond roundoff. These checks are not part of the proof, but guard against sign, facet-order, and reciprocal errors.

The theorem settles exactly the source’s conjecture for the open real hexablock. Formula (6) also extends continuously to the closed real tetrablock, but no claim is made here about convexity of the full complex hexablock or its other open problems.

A bounded novelty audit was performed on 26 August 2026. We searched the four run indexes, the local parsed arXiv corpus, arXiv/web results for the source id and title, the phrases “real hexablock” and “intersection of the hexablock,” and the radical/geometric-mean form in (5). We also inspected the abstracts of arXiv:2507.14589, arXiv:2507.16176, and arXiv:2608.00819, the later hexablock papers returned by the search. No prior convexity proof or this closed form was found. This is bounded evidence, not a definitive priority claim.

Human review recommendation: **send to human**, with priority on the half-angle simplification and the identification of the four facet functions.

References

- [1] I. Biswas, S. Pal, and N. Tomar, *The Hexablock: a domain associated with the μ -synthesis in $M_2(\mathbb{C})$* , arXiv:2506.15149v2, 2026.